

Multiple Choice (10 marks)

1. Imagine that the Earth was instantly moved to an orbit three times further away from the Sun. How much longer would a year be?
 - (a) exactly 3 times longer
 - (b) about 5.2 times longer
 - (c) Not enough information. It will depend on the inclination of the new orbit
 - (d) The length of the year wouldn't change because the Earth's mass stays the same.
2. Say the pupil of your eye has a diameter of 5 mm and you have a telescope with an aperture of 50 cm. How much more light can the telescope gather than your eye?
 - (a) 10000 times more
 - (b) 100 times more
 - (c) 1000 times more
 - (d) 10 times more
3. Which of these has NOT been one of the main hypotheses considered for the origin of the Moon?
 - (a) The Moon split from the Earth due to tidal forces.
 - (b) The Moon was captured into Earth orbit.
 - (c) The Earth and Moon co-accreted in the solar nebula.
 - (d) Earth was rotating so rapidly that the Moon split from it.
4. Which one of these constellations is not located along the Milky Way in the sky?
 - (a) Perseus
 - (b) Cygnus
 - (c) Scorpius
 - (d) Leo
5. In addition to the conditions required for any solar eclipse what must also be true in order for you to observe a total solar eclipse?
 - (a) The Earth must lie completely within the Moon's penumbra.
 - (b) The Moon's penumbra must touch the area where you are located.
 - (c) The Earth must be near aphelion in its orbit of the Sun.
 - (d) The Moon's umbra must touch the area where you are located.

True / False (10 marks)

Answer True or False

6. True or False: Water-ice is distributed evenly across Mercury, rather than being confined to the polar regions.
7. True or False: The Andromeda Galaxy is located approximately 2.5 million light years away from Earth.
8. True or False: The force required to accelerate the truck remains constant regardless of the planetary location due to uniform mass.
9. True or False: The resolution of a telescope is a measure of its ability to measure the angular separation of closely spaced objects.
10. True or False: The Moon's surface is less impacted by craters due to its thicker atmosphere compared to Earth's atmosphere.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons behind the varying compositions of inner and outer planets in our solar system, focusing on the role of temperature and material condensation in the protoplanetary nebula.
12. Discuss the concept of bolometric luminosity in astronomy, explaining its significance and the methods used to integrate luminosity across all wavelengths.

End of Paper